Daniel Hardesty Lewis

Education

2026 **M.S., Urban Planning**, *Columbia University*, New York City, NY
Thesis: *Predicting Formal Protest Petitions Against Housing Development: Evidence from Austin, Texas*

2017 **B.S., Pure Mathematics**, *University of Texas*, Austin, TX

2017 **Certificate, Scientific Computation**, *University of Texas*, Austin, TX

Experience

Professional

Dec. 2025 – Present **Founder**, *Worldcastr*, New York, NY
Spatiotemporal foundation model of the real world, and the platform that puts it behind high-stakes decisions: one model from the individual parcel or condo unit to the nation, and from days to decades.

- Trained FMTSFlow, a multi-target spatiotemporal forecaster, on a 388M-row panel of real public data across 16 jurisdictions in 4 countries and 32 targets (population, households, housing, rents, home values, income, employment, migration), one checkpoint per information cutoff, 2019 to 2023.
- Built a pre-registered benchmark that scores every target the same way: against published agency projections, a held-flat estimate and a gradient-boosted lag baseline, on later Census counts, 1 to 4 years ahead at tract and county, with county-clustered bootstrap intervals and an information-matching rule (an entry meets only projections published after its own cutoff).
- Across the scored targets (population, households, housing units) the model beats official projections in 6 of 10 pooled target-level comparisons and ties the rest; county population 59.8% win rate vs 18 sources (3,134 counties); tract 54.9% vs 30 agencies (households 60.1%, housing 52.5%, population 48.0%); the lag baseline still leads at tract; matched 2025 test under way.
- Assembled the publication-dated archive behind it (131 agency sources, 3,065 files, 27 normalized vintages) and packaged an arXiv preprint, Hugging Face dataset, GitHub scorer with leaderboard, and API model card for release (pending approval).
- Shipped the first product, Homecastr, a parcel-value diffusion forecaster (spatial inducing tokens, per-horizon loss normalization, DDIM sampling; PIT, coverage and sharpness checks before release) serving roughly 14M production requests weekly.

Jun. 2026 – Sep. 2026 **Member**, *South Park Commons*, New York, NY
Selective technology community for founders at a leading venture capital firm.

- Designed and implemented the next iteration of the Worldcastr machine learning architecture: a full-sequence inpainting editable world model.
- Conducted customer discovery with potential ICPs, including Parcl, Crusoe, and Cushman & Wakefield, to align predictive capabilities with enterprise requirements.

Oct. 2024 – Sep. 2025 **Technical Advisor**, *PoliBOM*, Seattle, WA
AI tariff mitigation and trade compliance product for manufacturers, centered on agentic BOM analysis and policy scenario simulation.

- Engineered an agentic retrieval-augmented pipeline for BOM parsing and tariff simulation, replacing manual trade-compliance analysis with predictable multi-step reasoning.
- Advised on the conversational AI interface enabling decision-makers to instantly retrieve BOMs and receive proactive compliance alerts.
- Executed customer discovery with ScoutyAI and Columbia economists, and secured early market validation by presenting the prototype to KPMG.

Nov. 2023 – Aug. 2025 **Founder**, *Summit Geospatial*, New York, NY
High-resolution elevation data and web delivery for engineering and hazard analysis: the highest quality seamless elevation data for Texas.

- Engineered a statewide seamless Texas DEM mosaic, resampling 0.5 m LiDAR sources to 1.2 m via nearest-neighbor across 70+ state, federal, and local elevation datasets.
- Developed the web distribution platform end-to-end, including data pipeline, tiling, hosting, and delivery UX, to support engineering and planning use cases.

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Aug. 2021 – Aug. 2023 **Senior Data Scientist & Technical Lead**, *Texas Advanced Computing Center (TACC)*, Austin, TX

- Spun out Summit Geospatial alongside UT's VPs of IP to commercialize state-level terrain models from the $40M Texas Disaster Information System (TDIS), delivering precision decision-support for civil engineering.
- Led a principal project of the $40M TDIS disaster resiliency initiative; developed inter-agency RFP and MOU frameworks with state and federal partners, translating technical requirements into scope and milestones facilitating Real-Time Inundation Mapping and TDIS.
- Scaled climate and flood models to system-scale, million-task jobs on Frontera (then a top-5 supercomputer, about 450K cores); managed $1M+ compute budgets and NOAA, NSF, USACE, and GLO partnerships.
- Developed efficient methods to produce high-resolution 1m flood maps from National Water Model outputs for Texas Emergency Management's Real-Time Inundation Mapping.
- Partnered with the US Army Corps of Engineers to statistically model compound flood hazards in coastal Texas, integrating high-resolution topographic data with hydrological models.
- Co-authored the inundation-mapping research presented to NOAA with Paola Passalacqua (2022 EGU Bagnold Medal).
- Served as Technical Reviewer for Frontera Doctoral Fellows (2023) and Large-Scale Computing Pathways (2022) applications, evaluating $10M+ in computing resource allocations on one of the world's largest supercomputers.
- Managed relationships and developed projects with federal and state agencies and universities including NOAA, NSF, US Army Corps of Engineers, Texas General Land Office, USC, UT, A&M.

Feb. 2018 – Jul. 2021 **Data Scientist & Research Engineer**, *Texas Advanced Computing Center (TACC)*, Austin, TX

- Competed in DARPA World Modelers to improve disaster resiliency and food security modeling in East Africa.
- Mentored Petrobras engineers in deep learning and HPC to enable institutional technology transfer.

Research

Sep. 2024 – May 2026 **Graduate Thesis Researcher**, *Columbia University GSAPP*, New York, NY

- Formulated S-IRM and I-DML (Invariant DML with ICP pre-filtering) to extract environment-invariant causal features and eliminate spatial wealth confounder bias; code.
- Designed an Adversarial Causal Spatial Autoencoder (PyTorch) using minimax gradient reversal to disentangle spatial opposition geometry from neighborhood wealth.

May 2025 – Aug. 2025 **Research assistant to Dir. Electrical Engineering Zoran Kostic**, *Columbia University*, New York, NY

Math benchmarking and finetuning of DeepSeek's GRPO algorithm

Jan. 2025 – Oct. 2025 **Research assistant to Dir. Financial Engineering Ali Hirsa (industry-sponsored research)**, *Columbia University*, New York, NY

- Developed a multi-asset CVAE latent factor model with Skew-T Mixture priors; achieved 85% R^2 vs. commercial SaaS (75%) and Fama-French (58%) on backtested holdout prediction of daily mutual fund returns.
- Derived a tractable, axiom-compliant per-feature SHAP algorithm using Hessian-based recursive clustering to provide stable temporal regime-dependent attributions for deep learning models.

July 2019 – Aug. 2019 **Summer research intern to Pres. AAAI Yolanda Gil**, *University of Southern California*, Los Angeles, CA

Integrated hydrological models into the MINT Platform, competing year-long with MITRE to provide DARPA the "Airflow for science"

Teaching

Jan. 2018 – Dec. 2020 **Co-instructor**, *The University of Texas at Austin*, Austin, TX

Helped design and teach graduate-level courses: *Machine Learning for the Geosciences* for Petrobras engineers, *Intelligent Systems for the Geosciences*, and *Scientific Computation*, including C++, CUDA, HPC, & optimization

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Key Projects

Homecastr chat engine	Built an LLM chat engine with 30+ function-calling tools over geographic data and comparables, with dual-model routing between Claude and GPT-4o Mini optimized for cost and latency.
Predicting Zoning Opposition	Developed, to our knowledge, the first property- and owner-level causal ML model of NIMBYism: Double Machine Learning (DML) over a 193k case-period panel to isolate the spatial tax of formal protest petitions on allowable building height, using 15+ years of individual homeowner protest data provided by the City of Austin (github.com/dhardestylewis/thesis)
M&A predictor	Built an M&A prediction model (Oct.–Dec. 2025) with Prof. Eric Talley on 1.5M Compustat firm-quarters; long-only portfolio beat the S&P 500 by 2%/yr (2004–2020), peaking at 6%/yr
Flood Hazard Assessment	Partnered with the US Army Corps of Engineers to statistically model compound flood hazards in coastal Texas, integrating high-resolution topographic data with hydrological models.
Texas Disaster Information System (TDIS)	Led development of the core elevation data layer for a web-based spatial data system that supports resilient decision-making at the state and local levels.
Real-Time Inundation Mapping	Developed computationally efficient methods to produce high-resolution (1m) flood inundation maps from National Water Model outputs for emergency response personnel.

Technology Skills

Modeling	PyTorch, diffusion and score-based models (DDIM), Transformers, LoRA/QLoRA/PEFT, scikit-learn, TensorFlow; calibration via PIT, coverage, sharpness, CRPS
Causal inference	Double ML, IRM, ICP, DoWhy, EconML, PyWhy, causal forests
LLM applications	function calling, RAG, embeddings, LangChain, LangGraph, DSPy, FAISS; Claude and OpenAI APIs
Data stack	Python, SQL, PL/pgSQL, Pandas, Polars, PyArrow, Parquet, PostgreSQL, PostGIS, Redis, Elasticsearch, Supabase
Geospatial	GDAL, GeoPandas, RasterIO, HAND, HEC-RAS, MODFLOW, National Water Model, LiDAR and DEM processing
HPC and infrastructure	Frontera, Stampede2 and Lonestar (TACC), Slurm, Launcher, CUDA, C++, Fortran, Singularity, Docker, Kubernetes, Modal, Ray, Spark, Dask, Airflow, AWS, GCP, Azure, CI/CD
Languages and tools	Bash, Perl, TypeScript, React, Git, GNU/Linux, macOS, Windows

Thesis

2026 *Hardesty Lewis, D.*. "Predicting Formal Protest Petitions Against Housing Development: Evidence from Austin, Texas". M.S. Thesis, Columbia University GSAPP. DOI: 10.7916/1swr-4871

Peer-reviewed articles

2021 July Gil, Y., D. Garijo, D. Khider, C. A. Knoblock, M. Osorio, H. Vargas, M. Pham, J. Pujara, B. Shbita, B. Vu, Y. Chiang, D. Feldman, Y. Lin, H. Song, V. Kumar, A. Khandelwal, M. Steinbach, K. Tayal, S. Xu, S. A. Pierce, L. Pearson, *D. Hardesty Lewis*, E. Deelman, R. F. da Silva, R. Mayani, A. R. Kemanian, Y. Shi, L. Leonard, S. .D. Peckham, M. Stoica, K. M. Cobourn, Z. Zhang, C. Duffy, L. Shu. "Artificial Intelligence for Modeling Complex Systems: Taming the Complexity of Expert Models to Improve Decision Making". The ACM Transactions on Interactive Intelligent Systems 11(2):11.

Conference papers

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2019 Mar. Garijo, D., J. Pujara, B. Vu, D. Feldman, R. Mayani, K. Cobourn, C. Duffy, A. R. Kemanian, L. Shu, V. Kumar, A. Khandelwal, D. Khider, K. Tayal, S. D. Peckham, M. Stoica, A. Dabrowski, *D. Hardesty Lewis*, S. A. Pierce, V. Ratnakar, Y. Gil, E. Deelman, R. F. da Silva, C. Knoblock, Y. Chiang, M. Pham. "An intelligent interface for integrating climate, hydrology, agriculture, and socioeconomic models". ACM 24th International Conference on Intelligent User Interfaces (IUI'19).

2018 June Garijo, D., D. Khider, Y. Gil, L. A. M. C. Carvalho, B. T. Essawy, S. A. Pierce, *D. Hardesty Lewis*, V. Ratnakar, S. D. Peckham, C. Duffy, J. L. Goodall. "A Semantic Model Catalog to Support Comparison and Reuse". 9th International Congress on Environmental Modelling and Software.

Technical reports

2021 Mar. Sun, A., M. H. Young, S. A. Pierce, J. Thompson, *D. Hardesty Lewis*, B. R. Scanlon. "Development of a Framework of Data Interpolation, Scaling, and Homogenization (DISH) for Mapping Natural Resources and Socioeconomic Data in Texas".

2019 Oct. Khider, D., Y. Gil, D. Garijo, K. M. Cobourn, C. Duffy, A. R. Kemanian, S. D. Peckham, B. Watkins, A. Campion, C. Preager, S. A. Pierce, *D. Hardesty Lewis*, A. Dabrowski, C. H. Porter, M. Landsfeld, M. Puma, B. Schauberger, A. Sliva, C. T. Morrison. "Towards a Shared Modeling Terminology and Problem Specification Framework".

Presentations

2021 May Passalacqua, P., F. R. Salas, R. Schomp, A. Carruthers, *D. Hardesty Lewis*. "Estimating Inundation Extent and Depth from National Water Model Outputs and High Resolution Topographic Data". Presented to the National Oceanographic and Atmospheric Administration.

2020 Dec. Sun, A., J. Thompson, *D. Hardesty Lewis*, J. Powell, M. H. Young, B. R. Scanlon, S. A. Pierce. "Development of a Framework of Data Interpolation, Scaling, and Homogenization (DISH) for Mapping Natural Resources in Texas". Presented at the 2020 annual Fall Research Showcase of Planet Texas 2050.

2020 Sep. Passalacqua, P., D. R. Maidment, D. Arctur, H. Evans, C. Thies, A. Carruthers, R. Schomp, *D. Hardesty Lewis*, S. A. Pierce. "From Rain Forecasts to Stream Flow to Flood Modelling". Presented at the 2020 annual Fall Research Showcase of Planet Texas 2050.

2020 Aug. *Hardesty Lewis, D.*. "Vector and Raster GIS Processing with Python in Jupyter Notebooks". Presented at the 2020 annual TACC Institute of Planet Texas 2050.

2019 Sep. *Hardesty Lewis, D.*, A. Dabrowski, J. Powell, S. A. Pierce. "DataX and MINT Overview". Presented at the 2019 annual Fall Research Showcase of Planet Texas 2050.

Posters

2019 Dec. Khider, D., Y. Gil, K. M. Cobourn, E. Deelman, C. Duffy, R. F. da Silva, A. R. Kemanian, C. A. Knoblock, V. Kumar, S. D. Peckham, Y. Chiang, D. Feldman, D. Garijo, *D. Hardesty Lewis*, A. Khandelwal, R. Mayani, M. Osorio, M. Pham, S. A. Pierce, J. Pujara, V. Ratnakar, L. Shu, H. J. Song, B. Shbita, M. Stoica, B. Vu, L. Pearson. "MINT: An intelligent interface for understanding the impacts of climate change on hydrological, agricultural and economic systems". Poster presented at the 2019 annual fall meeting of the American Geophysical Union.

2019 June *Hardesty Lewis, D.*, J. C.Thompson, E. Pease, Q. Yang, M. H. Young, S. A. Pierce. "A unified data representation of Texas water resources". Poster presented at the 2019 annual meeting of the Earthcube community.

2019 June Pierce, S. A., J. Powell, A. Karpatne, D. Garijo, J. Martin, *D. Hardesty Lewis*, P. Marchetto, S. Cleveland, M. Daniels, I. Athanasiadis, P. Keys, I. Demir, D. Fuka, S. Peckham, M. Hill, I. Ebert-Uphoff, D. Pennington, G. Jacobs, Y. Gil. "Intelligent Systems and Geosciences". Poster presented at the 2019 annual meeting of the Earthcube community.

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2019 June Powell J., A. Karpatne, D. Garijo, J. Martin, *D. Hardesty Lewis*, P. Marchetto, S. Cleveland, M. Daniels, I. N. Athanasiadis, P. W. Keys, I. Demir, S. D. Peckham, M. Hill, I. Ebert-Uphoff, D. Pennington, G. Jacobs, Y. Gil, S. A. Pierce. "Creating Sustainable Knowledge Centric Communities with Artificial Intelligence Applications to Earth Science Problems". Poster presented at the 2019 annual meeting of the Earthcube community.

2019 June Pease, E., J. C. Thompson, *D. Hardesty Lewis*, Q. Yang, S. A. Pierce, M. H. Young. "Integrated Modeling of Texas Water Resources". Poster presented at the 2019 annual meeting of the MODFLOW and More conference series.

2018 Dec. Pease, E., A. Pfeil, V. Ibarra, S. Siddique, F. Apango, O. Ramirez, E. Collado, *D. Hardesty Lewis*, N. Freed, S. A. Pierce. "Groundwater Modeling with Informatics and Automated Workflows for Water Resource Management: A Case Study from the Northern Trinity Aquifer". Poster presented at the 2018 annual fall meeting of the American Geophysicial Union.

2018 Sep. Martin, J., *D. Hardesty Lewis*, N. Freed, S. A. Pierce. "The IS-GEO Gateway: A community portal to facilitate AI and knowledge centered earth discoveries". Poster presented at the 13th annual conference of Gateway Computing Environments.

2018 June Martin, J., D. Garijo, N. Freed, S. A. Pierce, Y. Gil, D. R. Thompson, I. Demir, I. Ebert-Uphoff, D. Pennington, *D. Hardesty Lewis*, M. Hill, D. Fuka. "IS-GEO: A Research Coordination Network on Intelligent Systems Research to Support he Geosciences". Poster presented at the 2018 annual meeting of the EarthCube community.

2017 Dec. *D. Hardesty Lewis*, S. A. Pierce. "From MODFLOW-96 to MODFLOW-2005, ParFlow, and Others". Poster presented at the 2017 annual fall meeting of the American Geophysical Union.

2017 Dec. Kejriwal, M., S. A. Pierce, P. I. Q. Houser, S. D. Peckham, Z. Stanko, *D. Hardesty Lewis*. "Semi-automatic Data Integration using Karma". Poster presented at the 2017 annual fall meeting of the American Geophysical Union.

2016 July Cantu, A., S. A. Pierce, O. Rivera, A. Ramirez, *D. Hardesty Lewis*, J. Gentle, G. Fuentes-Pineda. "Big Data Analysis for Determining Sustainable Yield and Negotiation Space for an Aquifer System". Poster presented at the 51st annual meeting of the South-Central Section of the Geological Society of America.

2016 Apr. *Hardesty Lewis, D.* "On smoothness and Poisson's Equation". Presented at the 2016 annual DRP Undergraduate Project Symposium at the University of Texas at Austin.

Academic projects

2022 June – 2023 Aug. Advanced compound flood hazard assessment science
Partner with US Army Corps of Engineers to statistically model flood hazards in Texas's coastal region

2021 Oct. – 2023 Aug. Texas Disaster Information System (TDIS)
Provide analytical web-based spatial data system to support more resilient decision-making at the state level

2020 Dec. – 2023 Aug. Museum of South Texas History Sunday Speaker Series
Provide Dash-enabled website to facilitate geolocation of archival imagery by museum visitors

2020 Nov. – 2023 Aug. Networks for Hazard Preparedness and Response
Overlay new, finely tuned flood maps over specially designed building and social vulnerability maps

2020 Sep. – 2023 Aug. Real-time flood inundation mapping to improve community resilience
Provide simple, computationally efficient, high-resolution flood inundation maps to emergency response personnel

2020 Jan. – 2023 Aug. Estimating Inundation Extent and Depth from National Water Model Outputs and High Resolution Topographic Data
Improving the accuracy of flood inundation estimation products from 10m to 1m resolution

2017 Dec. – 2023 Aug. MINT: Model INTegration Through Knowledge-Rich Data and Process Composition
Integrate geoscience models MODFLOW, HAND, & SWAT with models from widely separate disciplines including agriculture, economics, and social sciences

2016 Aug. – 2023 Aug. Intelligent Systems for Geosciences
Used natural language processing and ETL pipelines to capture geoscientific variables' metadata to integrate with intelligent systems

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2019 Sep. – 2020 Aug. Improving the Estimation of Inundation Extent and Depth with High Resolution Terrain Data Over the State of Texas
Provided utilities to scale up the flood modelling toolset, GeoFlood, to statewide applicability across Texas

2018 July – 2019 Dec. Optimal Averaging of Water Resources in Texas
Indentified Texas water data sources, described their spatio-temporal scales and locations, estimated volume of all water in Texas its uncertainty, determined best methods to up-/down-scale all date to a common resolution and grid and minimise error

2017 May – 2017 Aug. From MODFLOW-96 to MODFLOW-2005, ParFlow, and Others
Brought a common groundwater flow model to scale on a supercomputer by updating its own Fortran code, using Perl natural language processing to extract and transform useful metadata from its user manual, and helping to get that metadata incorporated into a widely used scientific ontology

2016 July Big Data Analysis for Determining Sustainable Yield and Negotiation Space for an Aquifer System
Developed some Perl transformation routines between input file formats of groundwater models, MODFLOW-96 and ParFlow

2014 Oct. – 2014 Dec. Stochastic Differential Equations and Monte Carlo Simulation
Drafted Fortran routine that could run arbitrary equations of this type with arbitrary parameters

2013 May – 2013 Dec. Region of Calculated Existence for O/Ne-Core Stars with H/He Envelopes
Verified known observations, discovering that MESA accurately models this type of star under its default parameters

Professional projects

2019 Nov. – 2024 Dec. **Board member**, *Austin Community Reinvestment Cooperative (ACRC)*, Austin, United States
Helped establish cooperative real estate investment firm
Detailed accomplishments:
- Shaped bylaws and business plan, having consulted with advisors who've had 40+ years in cooperative development experience
- Provide GIS maps of specific properties to follow for a potential purchase, having selected them from criteria applied across parcel-level data
- Provide heat maps to help determine the best neighborhoods for investment based on parcel-level zoning, existing building characteristics, and simple Python pro formas
- Develop fully detailed real estate development pro formas for specific properties
- Develop and maintain relationships with Central Austin property owners interested in a collaboration
- Develop and submit options-to-purchase to interested property owners

2019 May – Present **Member-owner**, *Hardy Cooperative*, Austin, United States
Helped establish Central Austin affordable housing cooperative
Detailed accomplishments:
- Helped establish affordable housing cooperative of 7+ households in Central Austin, generating $60,000+ in annual revenue while providing rents 35% more affordable than nearby alternatives
- Collaborated with Dave Sullivan, former long-time chair of Austin planning commission, to determine that we met all zoning requirements for our use
- Secured 3 guarantors willing to front $40,000-$80,000 in liability for our commercial lease and new venture
- Drafted and implemented business prospectus, operational pro forma, resident lease applications, resident lease agreements, guarantor agreements, loan agreements, etc
- Maintained good relationships with 2 separate pro bono attorneys who have extensively reviewed, provided feedback, and largely rubber-stamped the contracts I have written
- Secured $30,000+ in lending, as well as conditional approval for another $40,000
- Received 150+ applicants for only 7 spaces in just 2 years of operation
- Established and maintained relationships with a 20+-year old housing coop and the ACRC in order to secure the long-term future of this venture

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2019 May – 2020 Apr. **Member, housing committee**, *Austin Cooperative Business Association*, Austin, United States
Austin Land Development Code (LDC) Revision
Detailed accomplishments:
- Reviewed cumulative 3000+ pages of 6 different Austin LDCs (1972, 1984, current, and LDC drafts 1, 2, & 3)
- Developed ACBA policy positions from understanding of these codes
- Distributed GIS maps depicting impact of each proposed LDC draft and our proposed LDC amendments on current and future affordable housing cooperative development
- Secured meeting with Mayor Steve Adler
- Advocated ACBA's position in direct meetings with Mayor Steve Adler, CM Greg Casar, CM Harper-Madison's policy director, CM Leslie Pool's chief of staff, and City Staff LDC drafters
- Notified co-ops impacted by potential code changes and put them in touch with relevant City Council and City Staff members
- Helped advance ACBA positions through different City Councilmember offices and conducted vote-counts of each position before and after each vote
- Succeeded in getting amendments adopted into draft LDC

Review committees

2023 Mar. – 2023 Apr. Frontera Fellows applications
Helped TACC determine the most qualified doctoral fellows' projects on one of the largest supercomputers in the world

2022 Aug. TACC's Frontera Pathways & Large-Scale Community Partnerships applications
Helped approved/deny millions of dollars in computing resources to multimillion-dollar projects

Students mentored

2021 Aug. – 2023 Aug. Khushboo Agarwal
MS Energy and Earth Resources, the University of Texas at Austin (expected 2023)

2020 Aug. – 2023 Aug. Mark Wang
PhD Civil Engineering, the University of Texas at Austin (expected 2025)
MS Environmental and Water Resources Engineering, the University of Texas at Austin (2022)

2019 Aug. – 2021 May Robert Schomp
MS Environmental and Water Resources Engineering, the University of Texas at Austin (2021)

2019 Aug. – 2020 May Alec Carruthers
MS Environmental Engineering Technology / Environmental Technology, the University of Texas at Austin (2020)

Pertinent coursework

- *Mathematics* Advanced Calculus for Applications, Linear Algebra and Matrix Theory, Differential Equations, Real Analysis I, Probability I, Partial Differential Equations and Applications, Topology I, Algebraic Structures I, Complex Analysis, Vector Calculus
- *Scientific computing* Introduction to Scientific & Technical Computing, Artificial Intelligence *(audited)*, Mathematical Modelling in Science and Engineering, Introduction to Stochastic Processes, Applied Statistics
- *Political and social sciences (2019 ICPSR Summer Program, University of Michigan)* Intro to Network Analysis, Machine Learning, Math for Social Sciences III

Professional affiliations

- *Mathematical Association of America* Member
- *North American Students of Cooperation* Member
- *American Geophysical Union* Member

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Extracurricular activities

- *Austin Transportation Equity Analysis Zones Advisory Team* Member, one of 30 members selected from a pool of 177 applicants to advise Austin Transportation on equitable geographic allocation of Project Connect's anti-displacement bond
- *Austin Community Reinvestment Cooperative (ACRC)* Board member
- *Austin Cooperative Business Assocation (ACBA)* Member, Development committee
- *Evolve Austin* Member, affiliate organization
- *AURA* Member
- *Brentwood Neighborhood Association* Member, provided PIR of full set of Affordability Unlocked applications
- *Hardy Cooperative* Founder, member-owner
- *Austin Cooperative Business Assocation (ACBA)* Member, Housing committee
- *Mission Interuniversitaire de Coordination Échanges Franco-Américains* Foreign exchange
- *Directed Reading Program* Graduate mathematical readings under individual guidance of doctoral students
- *Mathematics Club* Attended student- and professor-led talks
- *Inter-Cooperative Council (ICC) Austin* Member
- *College Houses Cooperatives* Member
- *Emerging Scholars Program, Calculus* Explored calculus outside of course material

Languages

French Professional working

Spanish Native

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